BOYCODE AFRICA — EXCEL PROJECT

One Dataset. Four Business Problems. Four Portfolio-Ready Projects.

Project Concept

All four houses will analyse the SAME raw dataset: a fictional Nigerian e-commerce company. The dataset contains 1,812 rows including intentional duplicate records and a small number of missing values. The exercise is designed to simulate the type of business data an analyst may receive from an operations, sales, marketing or management team.

The objective is not simply to demonstrate Excel functions. Each house must use Excel to answer a business problem, produce evidence, build a management-ready dashboard, and make recommendations that a real stakeholder could act on.

Common Dataset

- File: BoyCode_Africa_Shared_Ecommerce_Dataset.csv
- Rows: 1,812 raw records
- Core fields: orders, customers, regions, products, categories, revenue, cost, profit, marketing channel, delivery, payment, status and ratings.
- Data quality issues are intentionally included. Your first responsibility is to profile and clean the data.
- The data is synthetic and created for educational/portfolio purposes. It must be labelled as simulated data when published.

What a Hiring Manager Should See

- A clear business question rather than 'I analysed a dataset.'
- A documented cleaning process.
- Correct, traceable Excel calculations.
- A dashboard built for a decision-maker.
- Insights supported by numbers.
- Recommendations linked directly to the evidence.
- A short portfolio case study explaining the problem, approach, findings and actions.

KINGS HOUSE — SALES & PROFITABILITY PERFORMANCE REVIEW

You are a Business/Data Analyst supporting the Head of Commercial Operations. Management wants to understand what is driving revenue and profit and where the business should focus its sales effort.

Required Analysis:

1. Clean the raw dataset and document every issue you found.
2. Calculate total revenue, total cost, gross profit, gross margin %, total orders and units sold.
3. Analyse monthly revenue and profit trends.
4. Rank categories and products by revenue, profit and margin.
5. Identify products with high revenue but weak margin.
6. Analyse sales performance by region and customer segment.
7. Measure the effect of discounts on revenue and gross margin.

8. Identify the top 10 products that should receive more commercial attention and the bottom 10 that require investigation.
9. Build an executive sales dashboard with KPI cards, monthly trend, category performance, regional performance and a product profitability view.
10. Write a one-page management recommendation: What should the company sell more of, what should it review, and why?

ENERGY HOUSE — CUSTOMER SEGMENTATION & RETENTION ANALYSIS

You are a Customer Analytics Analyst. Management wants to understand who buys, how frequently they buy and which customer groups appear most valuable.

Required Analysis:

11. Create a customer-level summary from the transaction dataset.
12. Calculate each customer's total orders, total quantity, total revenue, total profit, average order value and average rating.
13. Classify customers into New, Returning and Loyal using a clear rule based on purchase frequency.
14. Identify the top customer segments by revenue and profit.
15. Analyse repeat-purchase behaviour by region and acquisition channel.
16. Calculate average order value and purchase frequency for customer groups.
17. Identify high-value customers and low-value/high-frequency customers.
18. Build a customer analytics dashboard showing customer count, repeat rate, average order value, revenue by segment, profit by segment, channel performance and regional customer value.
19. Recommend three practical actions to increase repeat purchases and customer value. Make clear which findings are descriptive rather than causal.

INTELLIGENCE HOUSE — MARKETING CHANNEL ROI & CUSTOMER ACQUISITION ANALYSIS

You are a Marketing Data Analyst. The marketing team spends money across several acquisition channels but does not know which channels generate the strongest commercial return.

Required Analysis:

20. Clean and validate the Acquisition_Channel and Marketing_Cost fields.
21. Calculate revenue, gross profit, marketing spend and net contribution after marketing cost by acquisition channel.
22. Calculate ROAS (Revenue / Marketing Cost) and marketing-cost-to-revenue %. Clearly document the formulas.
23. Compare channels by customer count, orders, average order value and profitability.
24. Analyse channel performance across regions and customer segments.
25. Identify channels that generate high revenue but relatively weak profit.
26. Identify channels that generate lower volume but strong efficiency.
27. Build a marketing performance dashboard showing spend, revenue, profit, ROAS, order volume and channel/region comparisons.
28. Recommend how management should prioritise, review or test its acquisition channels based on the evidence.
29. Add a short 'Caution' section explaining why the analysis cannot prove that a marketing channel caused a purchase.

LOVE & LIGHT HOUSE — OPERATIONS, DELIVERY & CUSTOMER EXPERIENCE ANALYSIS

You are an Operations Analyst. Management wants to understand delivery performance, cancellations, returns and customer satisfaction so that operational problems can be prioritised.

Required Analysis:

30. Clean and validate Delivery_Type, Promised_Days, Delivery_Days, Late_Delivery, Order_Status and Customer_Rating.
31. Calculate delivery-on-time rate, late-delivery rate, cancellation rate and return rate.
32. Analyse delivery performance by region, delivery type, category and customer segment.
33. Measure the relationship between late delivery and customer ratings using appropriate Excel summaries. Do not claim causation.
34. Identify regions with the highest operational risk.
35. Compare Express versus Standard delivery performance.
36. Analyse which categories or regions have high return/cancellation rates.
37. Build an operations dashboard showing total orders, on-time %, late %, cancellation %, return %, average delivery days and average rating.
38. Create a priority matrix identifying the operational areas that management should investigate first.
39. Provide five recommendations, each tied to a specific finding from the data.

Required Excel Techniques

- Excel Table and structured references
- Data cleaning and validation
- XLOOKUP or INDEX/MATCH where appropriate
- IF/IFS and logical functions
- SUMIFS, COUNTIFS, AVERAGEIFS
- Date functions and monthly grouping
- PivotTables and PivotCharts
- Conditional formatting
- Charts selected for a business purpose
- Dashboard layout and KPI cards
- Basic error handling and validation checks

Final Deliverables

40. 1. Cleaned Excel workbook (.xlsx)
41. 2. Raw Data sheet preserved separately from the cleaned dataset
42. 3. Data Cleaning / Quality Check sheet
43. 4. Analysis sheet(s)
44. 5. PivotTables/PivotCharts
45. 6. One-page management dashboard
46. 7. One-page Insight & Recommendation summary
47. 8. Portfolio README/case-study summary (maximum 1 page)

Portfolio Case Study Template

- Business Problem — What decision did the business need to make?

- Dataset — What does the dataset contain?
- Data Preparation — What problems did you find and how did you fix them?
- Analysis — What methods and Excel techniques did you use?
- Key Findings — What are the 5 most important numbers or patterns?
- Recommendations — What should management do next?
- Limitations — What cannot be concluded from this dataset?
- Tools — Microsoft Excel

Assessment — 100 Marks

Area	Marks
Data cleaning & quality control	15
Excel formulas & calculations	20
Analytical reasoning	20
PivotTables/charts	10
Dashboard & visual communication	15
Insights & recommendations	15
Documentation & presentation	5

Submission Deadline

Monday, 31 August 2026 — 11:59 PM.

Submit the completed workbook and portfolio case study to the designated BoyCode Africa assignment Google Drive folder.

Professional Rule

You may use AI to understand formulas, troubleshoot errors, or improve presentation, but you must understand every calculation and recommendation in your submission. During presentation, any house member may be asked to explain any part of the workbook.